



PROJECT DOCUMENTATION

E-Commerce Order & Customer Analytics Platform

Advanced Excel Project Report

Group Member Details

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Project Details

Project Title	E-Commerce Order & Customer Analytics Platform
Project Duration	4 Days
Trainer Name	Prof. Sayan Sau
Submission Date	29-08-2026
GitHub Username	Sudeep0069
GitHub Repository Link	https://github.com/Sudeep0069/E-Commerce-Order-Customer-Analytics-Platform.git

1. Project Objective

The E-Commerce Order & Customer Analytics Platform was built to give business stakeholders a single source of truth for order, customer and product performance. It consolidates raw transactional and customer data into one governed Excel workbook that supports revenue tracking, profitability analysis and customer segmentation, replacing scattered raw exports with a structured, formula-driven model that can be refreshed, audited and extended over time.

2. Tools & Technologies Used

- Microsoft Excel — core workbook, formulas and calculated fields
- Pivot Tables — for KPI aggregation and dashboard views
- Excel Functions — SUM, COUNT, AVERAGE, and Index Match/join-style formulas used to merge Sales with Products and Customers
- Charts & Dashboard visuals — pivot-chart based visual dashboard
- Data Validation & layered sheet structure — to separate raw data from calculated and reporting layers

3. Dataset Description

The dataset spans June 2024 to June 2026 and covers 250,000 orders, 40,000 registered customers and 2,000 products across 7 categories and 33 brands, spanning 39 cities in 10 Indian states. It is organized into three base entity tables:

- Customers (40,000 rows) — demographics, location, registration date, loyalty tier, lifetime orders and spend
- Products (2,000 rows) — category, brand, pricing waterfall, stock and ratings
- Sales (250,000 rows) — order-level transactions including pricing, payment mode, delivery status and ratings

These are joined into a Master Data table (250,000 rows) enriched with cost price and a calculated profit field, which in turn feeds the KPI and Dashboard reporting layers.

4. Data Architecture & Methodology

4.1 Sheet Layers

Sheet	Records	Purpose
Customers	40,000 rows	Customer master — demographics, location, registration & loyalty tier
Products	2,000 rows	Product catalog — category, brand, pricing, stock & ratings
Sales	250,000 rows	Order-level transactions — pricing, payment, status & ratings

Sheet	Records	Purpose
Master Data	250,000 rows	Sales enriched with cost price and calculated profit per order
KPIs	—	Total Revenue, Total Profit, Average rating, Order Cancellation etc.
Dashboard	—	Pivot-table based views feeding the visual dashboard

4.2 Data Flow

1. Raw records are captured in the Customers, Products and Sales sheets.
2. Sales records are joined to Products (for price & details) and Customers (for Name & Personal Info) to build Master Data.
3. Profit is calculated per order in Master Data as Total Amount minus Cost Price × Quantity.
4. KPIs and Dashboard sheets summarize Master Data and Customers using pivot tables and formulas.

4.3 Segmentation Logic

- Customer age groups follow fixed bands: 18–25, 26–35, 36–45, 46–55, 56–65, and 65+.
- Loyalty tier (Silver / Gold / Platinum) is assigned per customer based on historical spend and order behavior.
- Geographic analysis is rolled up from order-level City and State fields to identify regional demand concentration.

5. Key Performance Indicators

KPI	Value
Total Orders	250,000
Total Customers	40,000
Total Products	2,000 (7 categories, 33 brands)
Total Revenue (Total Amount)	₹593.1 Cr
Total Profit	₹48.6 Cr (8.2% margin)
Average Product Rating	4.4 / 5 (528,060 reviews)
Average Stock per Product	735 units
Cities / States Covered	39 cities / 10 states
Most Common Payment Mode	UPI (51.4% of orders)
Most Common Product Category	Electronics

6. Key Features

- Layered workbook design separating raw entity tables, a merged Master Data layer, and reporting layers (KPIs, Dashboard)
- Order-level profitability calculation, not just aggregate revenue tracking
- Customer segmentation by loyalty tier, age group and geography for targeted marketing
- Payment-method and order-fulfillment-status monitoring for operational planning
- Pivot-table driven dashboard with standardized KPIs that update automatically as new data is added

7. Analysis & Insights

7.1 Revenue by Category & Brand

Electronics is by far the largest revenue contributor at ₹430.6 Cr (72.6% of total revenue), followed by Home (₹71.6 Cr, 12.1%) and Sports (₹47.6 Cr, 8.0%). Fashion, Grocery, Beauty and Books together make up the remaining 6.6%. At the brand level, HP (₹94.2 Cr), Noise (₹91.4 Cr), boAt (₹84.2 Cr), Samsung (₹83.5 Cr) and Apple (₹77.2 Cr) are the top five performers by revenue.

7.2 Customer Profile

Orders are concentrated in younger adult segments: the 26–35 age group places 35.1% of all orders (87,847 orders), followed by 18–25 at 24.6% (61,594 orders). Older age bands (56–65 and 65+) together account for only 8.0% of orders. By loyalty tier, Platinum customers make up 68.4% of the 40,000-customer base and contribute the vast majority of lifetime spend, while Gold (16.7%) and Silver (14.9%) customers contribute comparatively little. The customer base is evenly split by gender (50.0% male, 50.0% female).

7.3 Geographic Distribution

Revenue is broad-based across India rather than concentrated in one region. Uttar Pradesh (₹76.8 Cr), Rajasthan (₹75.7 Cr) and Haryana (₹75.5 Cr) are the top three states by revenue, with Maharashtra, Punjab and Gujarat each contributing roughly ₹60 Cr. Order activity spans 39 distinct cities in total.

7.4 Payment Behavior

UPI is the clear preference among customers, used in 51.4% of the 250,000 orders (128,474 orders), ahead of Cash on Delivery at 32.8% (82,093 orders). Card payments — Debit and Credit combined — account for the remaining 15.7% of orders, suggesting continued room to grow card-based checkout adoption.

7.5 Order Fulfillment

Of all orders placed, 80.1% (200,139) were successfully Delivered. The remaining 19.9% is split evenly across Shipped (5.0%), Returned (5.0%), Processing (5.0%) and Cancelled (5.0%) — each representing roughly 12,400–12,500 orders. The similar size of the Returned and Cancelled buckets is worth further investigation to identify recurring causes.

7.6 Revenue Trend

Monthly revenue has been highly stable across the full 25-month window (June 2024 – June 2026), fluctuating narrowly between ₹21.5 Cr and ₹24.8 Cr per month with no clear seasonal collapse or runaway growth — indicating a mature, steady-state order volume rather than an early-growth or declining trend.

8. Recommendations

5. Diversify category mix: Electronics' 72.6% revenue share creates concentration risk; invest in promoting Home, Sports and Fashion to balance the portfolio.
6. Protect the Platinum segment: with 68.4% of customers and the majority of lifetime spend concentrated in this tier, prioritize retention programs (early access, loyalty rewards) for this group.
7. Optimize for UPI-first checkout: given 51.4% adoption, ensure UPI remains the fastest, most reliable payment path, while continuing to support COD for reach in emerging areas.
8. Investigate returns and cancellations: the 5.0% Returned and 5.0% Cancelled rates are worth root-cause analysis — product-quality issues, delivery delays or checkout friction may be contributing factors.
9. Target growth segments: with 26–45-year-olds driving the majority of orders, tailor marketing and product assortment to this demographic while testing campaigns to grow the underrepresented 46+ segments.

9. Learning Outcome

This project taught us to merge multi-table data using lookup formulas, build calculated fields like profit, create pivot-table dashboards, and translate KPIs into real business insights and recommendations.

10. Conclusion

The E-Commerce Order & Customer Analytics Platform successfully brings together customer, product and sales data into a single analytical model capable of answering core business questions around revenue, profitability, customer value and operational performance. With ₹593.1 crore in revenue and an 8.2% profit margin across 250,000 orders, the underlying business is healthy and stable. The clearest opportunities going forward are diversifying beyond Electronics, deepening engagement with the high-value Platinum customer segment, and reducing the combined 10% of orders lost to returns and cancellations.

13. Trainer Verification Section

Trainer Remarks	
Signature	